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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

CS3332-1 – NATURAL LANGUAGE PROCESSING
(For CSE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **A Non-deterministic FSA (NFSA) differs from a deterministic one because**
A) It can process infinite strings B) It can move between multiple states for the same input symbol
C) It cannot accept any string D) It is faster computationally
2. **The minimum edit distance between the words 'kitten' and 'sitting' is**
A) 2 B) 3
C) 4 D) 5
3. **An un-smoothed bigram model has the major disadvantage of**
A) Overestimating rare words B) Being computationally expensive
C) Assigning zero probability to unseen word pairs D) Failing to count word frequencies
4. **Which type of morphology creates new words by adding prefixes or suffixes?**
A) Derivational B) Cliticization
C) Inflectional D) Non-concatenative
5. **Agreement in morphology refers to**
A) Matching tense B) Matching grammatical features like number and gender
C) Combining words D) Removing affixes
6. **In language modeling, the unigram assumption is that**
A) Words depend on the last two words B) Words depend only on the immediately previous word
C) Words depend on context-free grammar D) Words are independent of each other
7. **In HMM-based POS tagging, the transition probabilities represent**
A) Probability of a word given a tag B) Probability of a word given the previous word
C) Probability of a tag given the previous tag D) Probability of an edit distance
8. **In NLP, the n-gram model is primarily used for:**
A) Predicting the next word in a sequence B) Part-of-speech tagging
C) Parsing syntax trees D) Speech-to-text conversion
9. **A context-free grammar (CFG) consists of**
A) Terminals, Non-terminals, Productions, and Start symbol B) Alphabet, States, Start symbol, Accept states
C) Terminals, States, Transitions D) Semantics and Production Rules
10. **A bottom-up chart parser is preferred because**
A) It only uses breadth-first strategy B) It always guarantees shortest parse
C) It avoids recomputation of sub-parses D) It works only for deterministic grammars

11. First-Order Logic (FOL) allows us to represent
 A) Only syntactic categories B) Objects, Properties, and Relations
 C) Only probabilities of events D) Morphological variations
12. Which parsing strategy explores the left-most derivation first?
 A) Top-down parsing B) Bottom-up parsing
 C) Chart parsing D) Shift-reduce parsing
13. Which representation correctly models the sentence 'Some students failed the exam' in FOL?
 A) $\forall x(\text{Student}(x) \rightarrow \text{Failed}(x))$ B) $\exists x(\text{Student}(x) \wedge \text{Failed}(x))$
 C) $\exists x(\text{Failed}(x) \rightarrow \text{Student}(x))$ D) $\forall x(\text{Failed}(x) \wedge \text{Student}(x))$
14. Semantic augmentation to CFG rules is used to
 A) Increase parsing speed B) Reduce grammar size
 C) Provide meaning composition along with syntax D) Handle only phonological constraints
15. Which is a key difference between compositional and non-compositional semantics?
 A) Non-compositional semantics always uses probability B) Non-compositional semantics cannot handle idioms
 C) Compositional semantics ignores syntax D) Compositional semantics builds meaning from parts, non-compositional does not
16. Which is a challenge of computational semantics?
 A) Handling world knowledge and context B) Defining syntax rules
 C) Detecting spelling errors D) Counting corpus words
17. NER is often treated as what type of problem?
 A) Sequence labeling B) Parsing
 C) Topic modeling D) Semantic parsing
18. In Relation Extraction, the task is to
 A) Detect word senses in context B) Classify text into categories
 C) Generate natural language summaries D) Find semantic relationships between entities
19. Which of the following is a common entity type in NER tasks?
 A) Adjectives B) Pronouns
 C) Organizations D) Prepositions
20. The Factoid QA task aims to
 A) Generate detailed paragraphs B) Retrieve exact factual answers like names or dates
 C) Summarize multiple documents D) Evaluate coherence

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Discuss the role of Non-Deterministic FSA in recognizing sheep-talk. Explain an algorithm for NFSA recognition.	8	L*2	2	1
	b) Explain the Minimum Edit Distance algorithm and draw the minimum edit distance table for the string "Kite" to Kitten".	8	L2	2	1
2.	a) What is Regular Expression? Explain the basic regular expression patterns with suitable examples.	8	L2	1	1
	b) Define N-gram models. Show bigram probability with an example.	8	L2	2	1
3.	a) Define Morpheme. Explain Inflectional and derivational morphology.	8	L2	1	1
	b) Define the Markov chain. Explain the components in the Hidden Markov Model with suitable examples.	8	L2	2	1
Unit – II					
4.	a) Explain the Top-Down Chart parsing with an algorithm	8	L2	3	1
	b) Explain the First Order Logic with an example.	8	L2	4	1

5. a) construct the top-down parse tree for the string "The old man cried".
Use the following grammar.
S -> NP VP
NP -> ART ADJ N
NP -> ART N
VP -> V
VP -> V NP
- | | | | | |
|---|---|----|---|---|
| | 8 | L3 | 3 | 1 |
| b) Explain quantifier scope ambiguity and how under-specification helps resolve it. | 8 | L2 | 4 | 1 |
| 6. a) Explain computational desiderata for semantic representations. | 8 | L2 | 4 | 1 |
| b) Explain unification-based approaches to semantic analysis. | 8 | L2 | 4 | 1 |

Unit – III

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|---|---|----|---|---|
| 7. a) Explain the Basic steps in the statistical sequence labelling approach to creating a named entity recognition system. | 8 | L2 | 5 | 1 |
| b) Explain the basic architecture of a generic single document summarizer. | 8 | L2 | 5 | 1 |
| 8. a) Explain the properties of human conversations. | 8 | L2 | 5 | 1 |
| b) Explain the basic dialogue system with architecture diagram. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
